FUSING APACHE SPARK AND LUCENE FOR NEAR-REALTIME PREDICTIVE MODEL BUILDING

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verizon

Data Overview

- Location data
  - Each srcIp defined as unique row key
  - Provides approximate location of each key
  - Timeseries containing latitude, longitude, error bound, duration, timezone for each key
- Clickstream data
  - Contains clickstream data of each row key
  - Contains startTime, duration, httphost, httpuri, upload/download bytes, httpmethod
  - Compatible with IPFIX/Netflow formats

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Marketing Analytics

- Anonymous aggregate analysis for customer insights

+ Lookalike modeling

= Churn reduction

= Competitive analysis

+ Increased share of stomach

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Data Model

- Dense dimension, dense measure
  Schema: srcip, date, hour, tld, zip, tldvisits, zipvisits
  Data: 10.1.13.120, d1, H2, macys.com, 94555, 2, 4
- Sparse dimension, dense measure
  Schema: srcip, date, tld, zip, clickstreamvisits, zipvisits
  Data: 10.1.13.120, d1, {macys.com, kohls.com}, {94555, 94301}, 10, 15
- Sparse dimension, sparse measure
  Schema: srcip, date, tld, zip, tldvisits, zipvisits
  Data: 10.1.13.120, d1, {macys.com, kohls.com}, {94555, 94301}, {macys.com:4, kohls.com:6}, {94555:8, 94301:7}
  Schema: srcip, week, tld, zip, tldvisits, zipvisits
  Data: 10.1.13.120, week1, {macys.com, kohls.com}, {94555, 94301}, {macys.com:4, kohls.com:6}, {94555:8, 94301:7}
  - Sparse dimension, sparse measure, **last N days**
    Schema: srcip, tld, zip, tldvisits, zipvisits
    Data: 10.1.13.120, {macys.com, kohls.com}, {94555, 94301}, {macys.com:4, kohls.com:6}, {94555:8, 94301:7}
- Competing technologies: PowerDrill, Druid, LinkedIn Pinot, EssBase

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Document Dataset Representation

- Example
  Schema: srcip, tld, zip, tldvisits, zipvisits
  Data: 10.1.13.120, {macys.com, kohls.com}, {94555, 94301}, {macys.com:4, kohls.com:6}, {94555:8, 94301:7}
- DataFrame row to Lucene Document mapping

Store/schema	*Row*	*Document*
srcip	*primary key*	*docId*
tld *zip*	*String* *Array[String]*	*SingleValue/MultiValue* *Indexed Fields*
tldvisits *zipvisits*	*Double* *Map[String, Double]*	*SparseVector* *StoredField*

- Distributed collection of srcIp as RDD[Document]
  - ~100M srcip, 1M+ terms (sparse dimensions)

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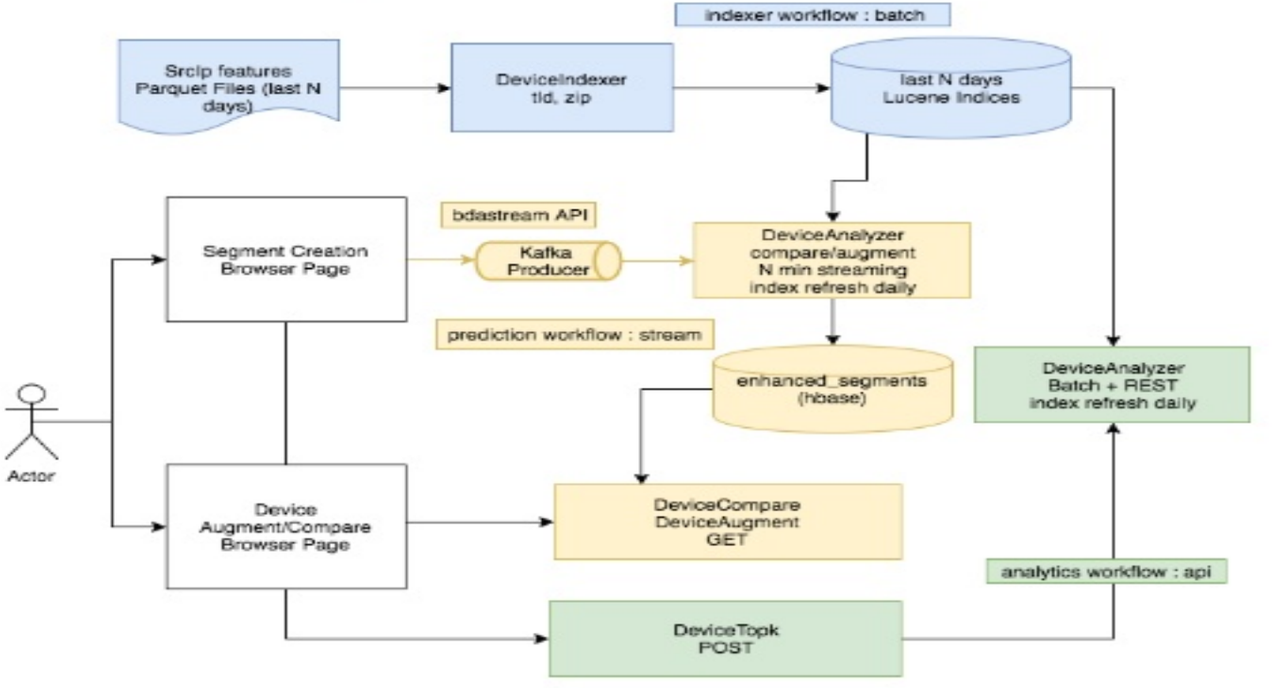
DeviceAnalyzer

- DeviceAnalyzer goals
  - Search and retrieve devices that matched query
  - Generate statistical and predictive models on retrieved devices



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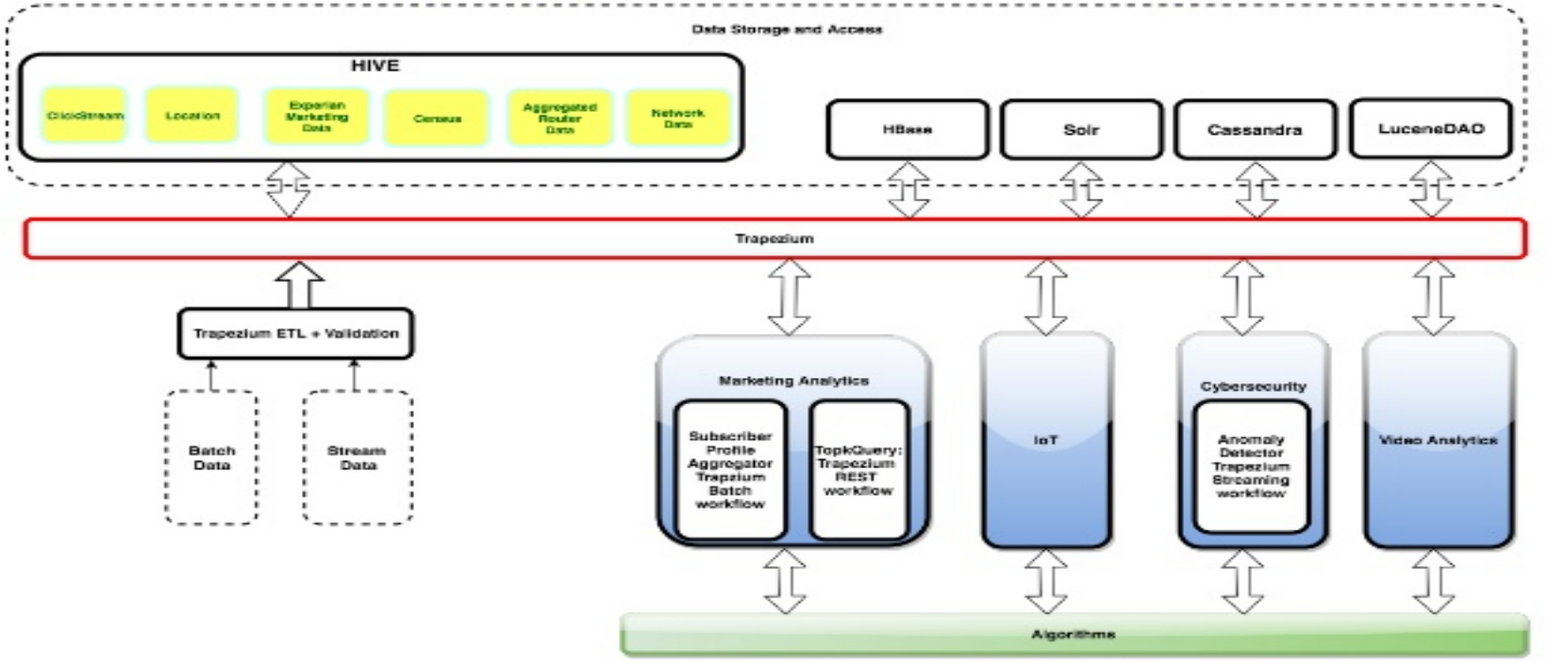
What is Trapezium ?

DAIS Open Source framework to build batch, streaming and API services

https://github.com/Verizon/trapezium



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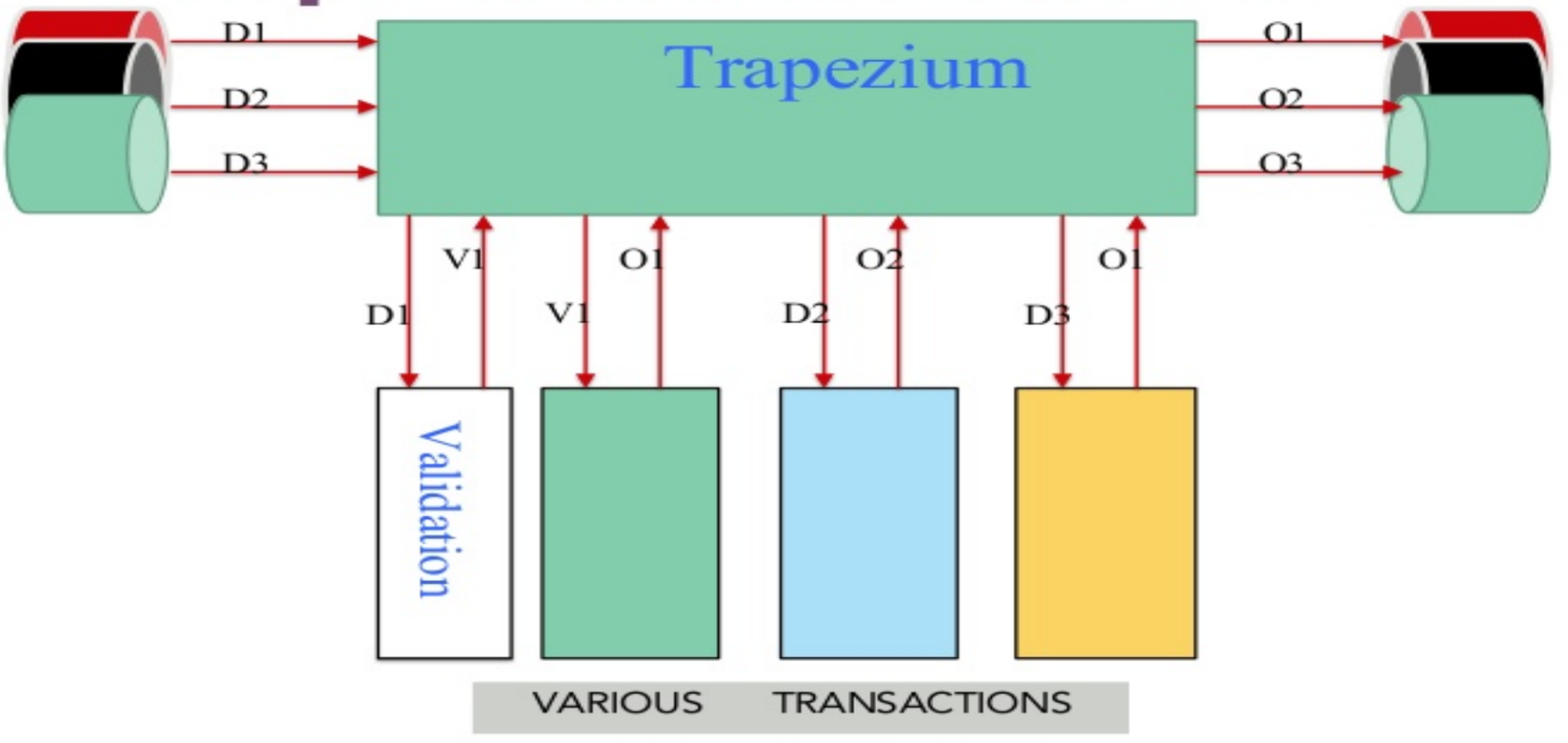
Trapezium Architecture



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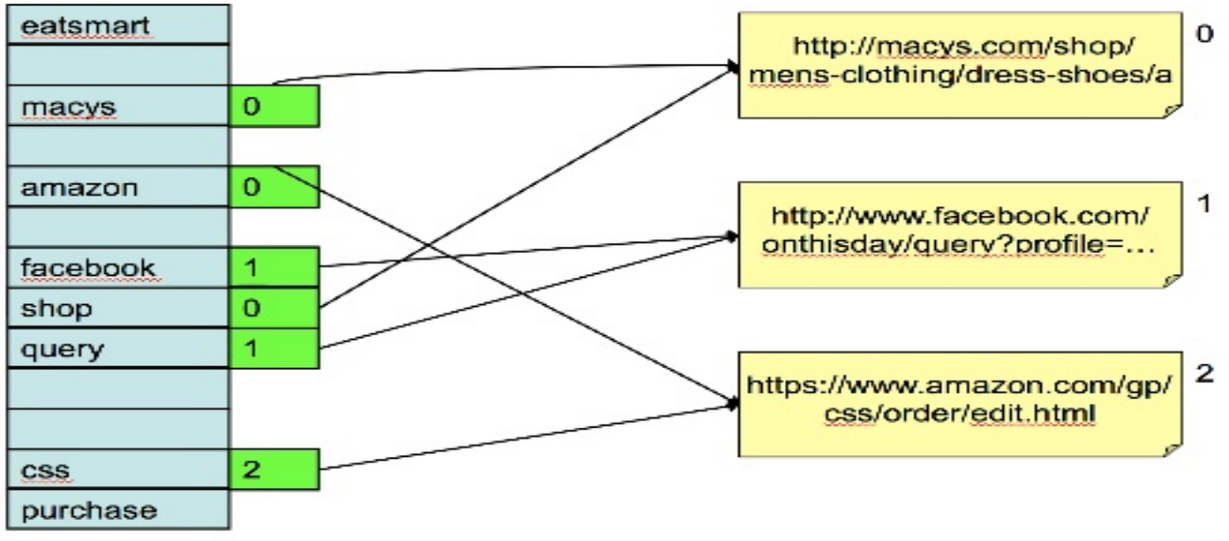
Lucene Overview

- Scalable, full-text search library
- Focus: Indexing + searching documents



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Trapezium LuceneDAO

- SparkSQL and MLlib optimized for full scan, **column indexing not supported**
- Why Spark + Lucene integration
  - Lucene is battle tested Apache Licensed Open Source Project
  - Adds column search capabilities to Spark
  - Adds spark operators (treeAggregate, treeReduce, map) to Lucene
- LuceneDAO features
  - Build distributed lucene shards from Dataframe
  - Save shards to HDFS for QueryProcessor (CloudSolr)
  - Access saved shards through LuceneDAO for ML pipelines

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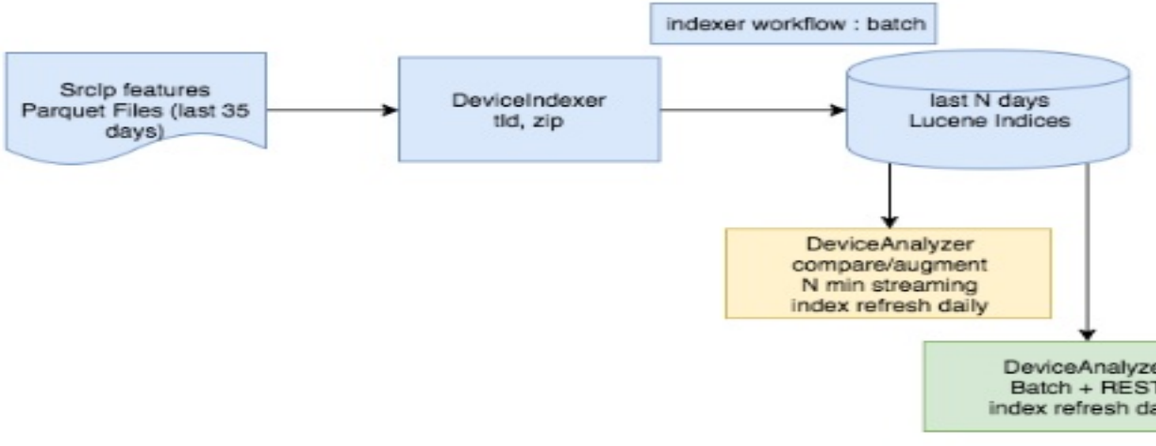
Trapezium Batch

```
runMode = "BATCH"
dataSource = “HDFS”
dependentWorkflows = {
    workflows=[aggregate]
    frequencyToCheck=100
}
hdfsFileBatch = {
  batchTime = 86400
  timerStartDelay = 1
  batchInfo = [{
      name = "DeviceStore"
      dataDirectory = {saiph-devqa=/aggregates}
      fileFormat = "parquet"
    }]
}
```

```
transactions = [{
  transactionName = “DeviceIndexer”
  inputData = [{name = "DeviceStore"}]
  persistDataName = “indexed"
}]
```



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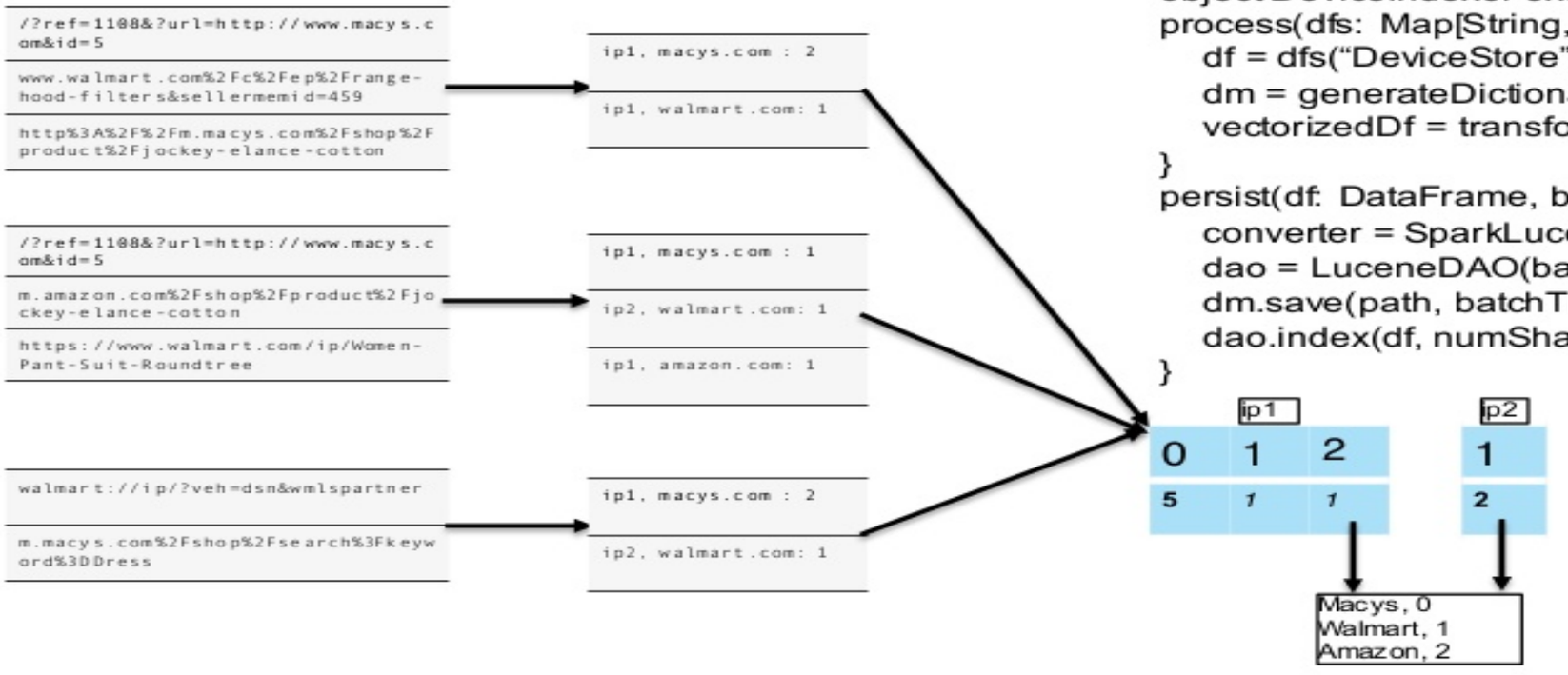
DeviceAnalyzer: Indexing




```
object DeviceIndexer extend BatchTransaction {
process(dfs: Map[String, DataFrame], batchTime: Time): {
    df = dfs("DeviceStore")
    dm = generateDictionary(df)
    vectorizedDf = transform(df, dm)
}
persist(df: DataFrame, batchTime: Time): {
    converter = SparkLuceneConverter(dm.size)
    dao = LuceneDAO(batchTime,...).setConverter(converter)
    dm.save(path, batchTime)
    dao.index(df, numShards)
}
```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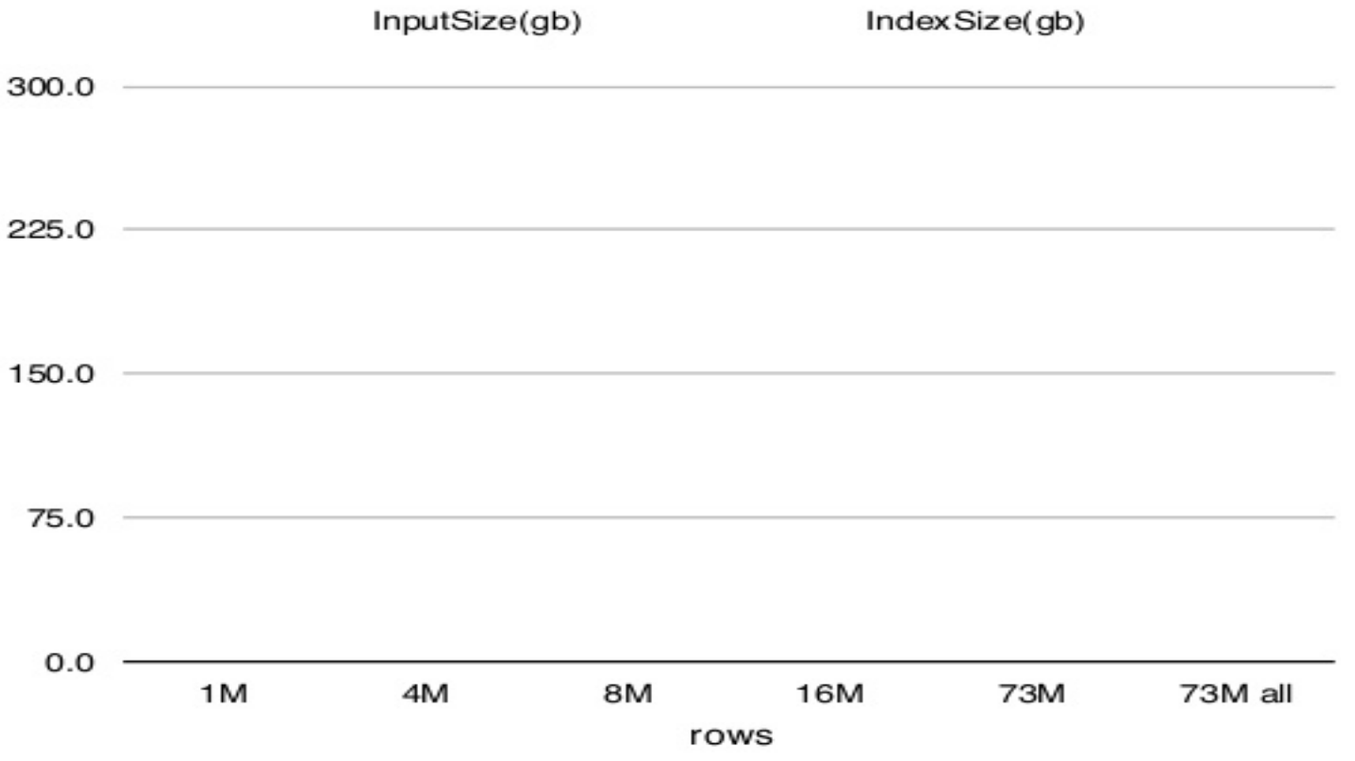
LuceneDAO Index Size

rows	InputSize(gb)	IndexSize(gb)
1M	4.0	5.1
4M	14.4	19.0
8M	27.9	35.7
16M	58.8	63.2
73M	276.5	228.0
73M all	276.5	267.1



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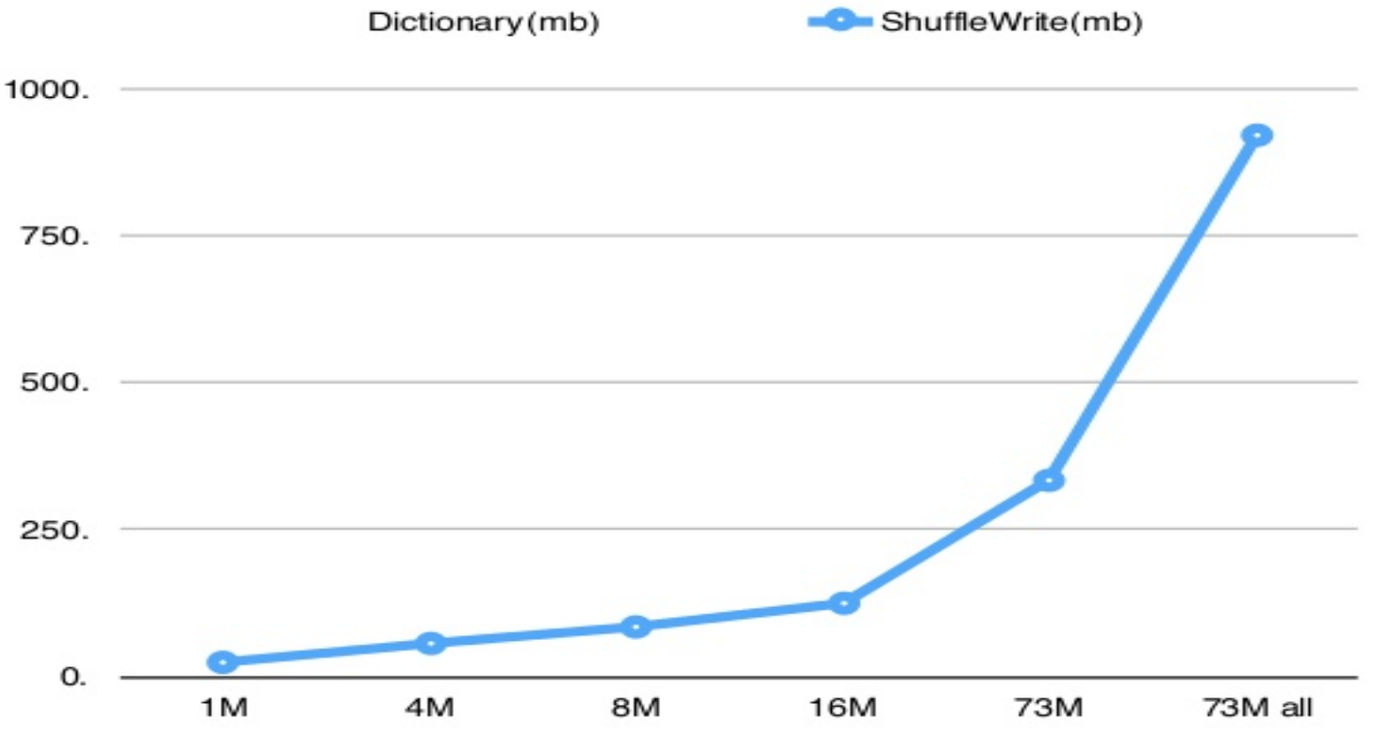
LuceneDAO Shuffle Size

rows	ShuffleWrite(mb)	Dictionary(mb)
1M	25	22.0
4M	56	30.0
8M	85	31.6
16M	126	32.2
73M	334	32.4
73M all	921	146.5



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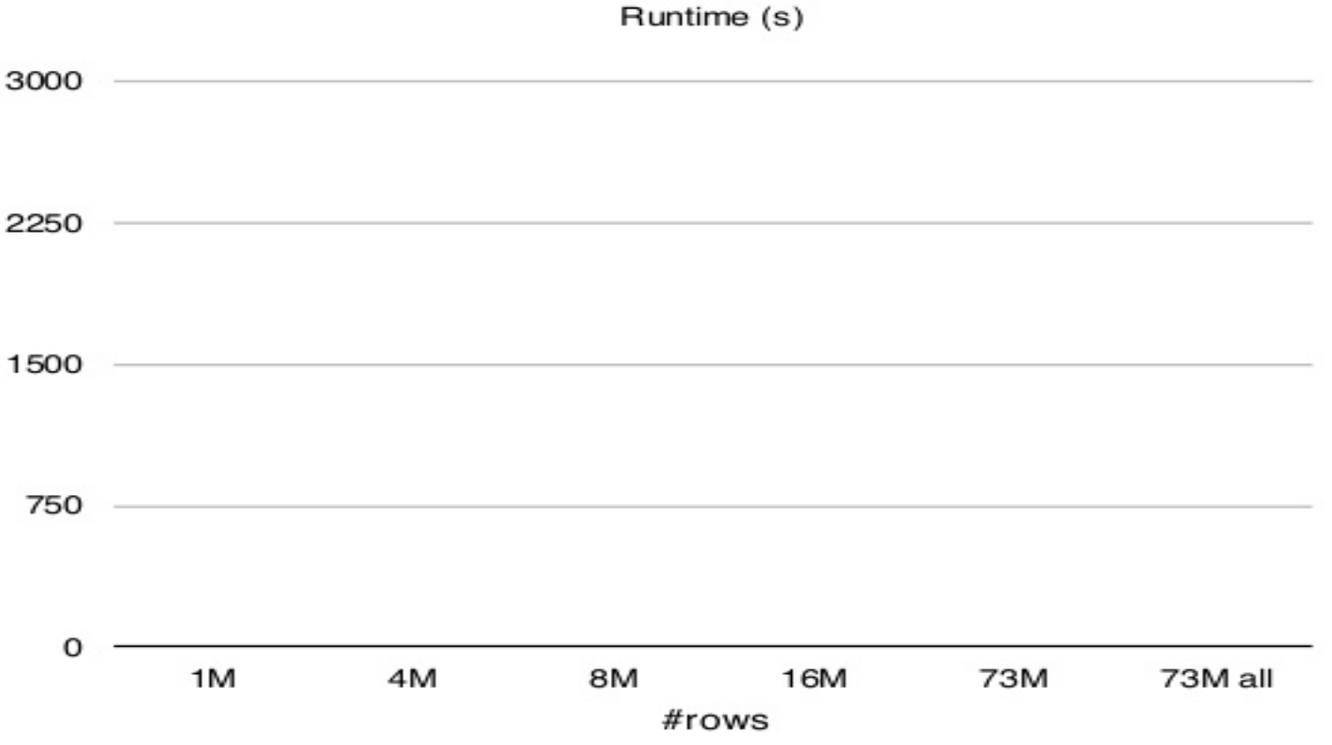
LuceneDAO Index Runtime

20 executors 16 cores
Executor RAM 16 GB
Driver RAM 8g

rows	Runtime (s)
1M	135
4M	228
8M	434
16M	571
73M	1726
73M all	2456



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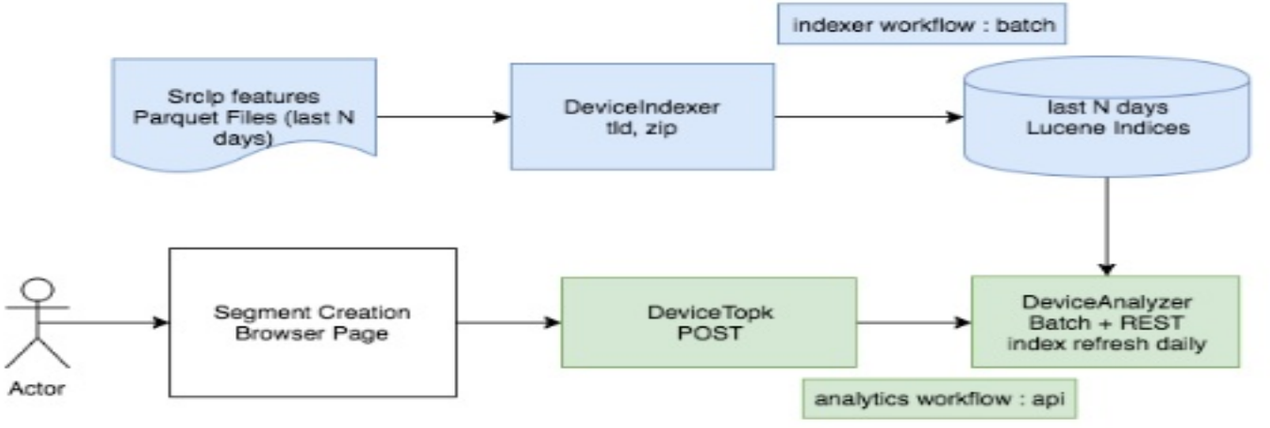
Trapezium Api

```
runMode = "BATCH"
dataSource = "HDFS"
httpServer = {
  provider = "akka"
  hostname = "localhost"
  port = 19999
  contextPath = "/"
  endPoints = [{
      path = “analyzer-api"
      className = "TopKEndPoint"
    }]
}
```



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DeviceAnalyzer: Topk

- Given a query select * from devices where tld='macys.com' OR 'nordstorm.com' AND (city='SanFrancisco' OR 'Brussels') AND (device='Android') ...
  - ML: Find topk dimensions highly correlated with selected device
  - BI: group by tld order by sum(visits) as tldVisits limit topk

```
class TopkController(sc: SparkContext) extends
SparkServiceEndPoint(sc) {
override def route : topkRoute
converter = SparkLuceneConverter(dm.size)
batchTime = Trapezium.getSyncTime("indexer")
dao = LuceneDAO(batchTime...)
    .setConverter(converter).load(sc, indexPath)
dict = loadDictionary(sc, indexPath, batchTime)
def topkRoute : {
  post { request => {
    devices = dao.search(request)
    response = getCorrelates(devices, dict, topk)
  }
}
```

df[deviceId, vector]

sum, support
mean, median, stddev

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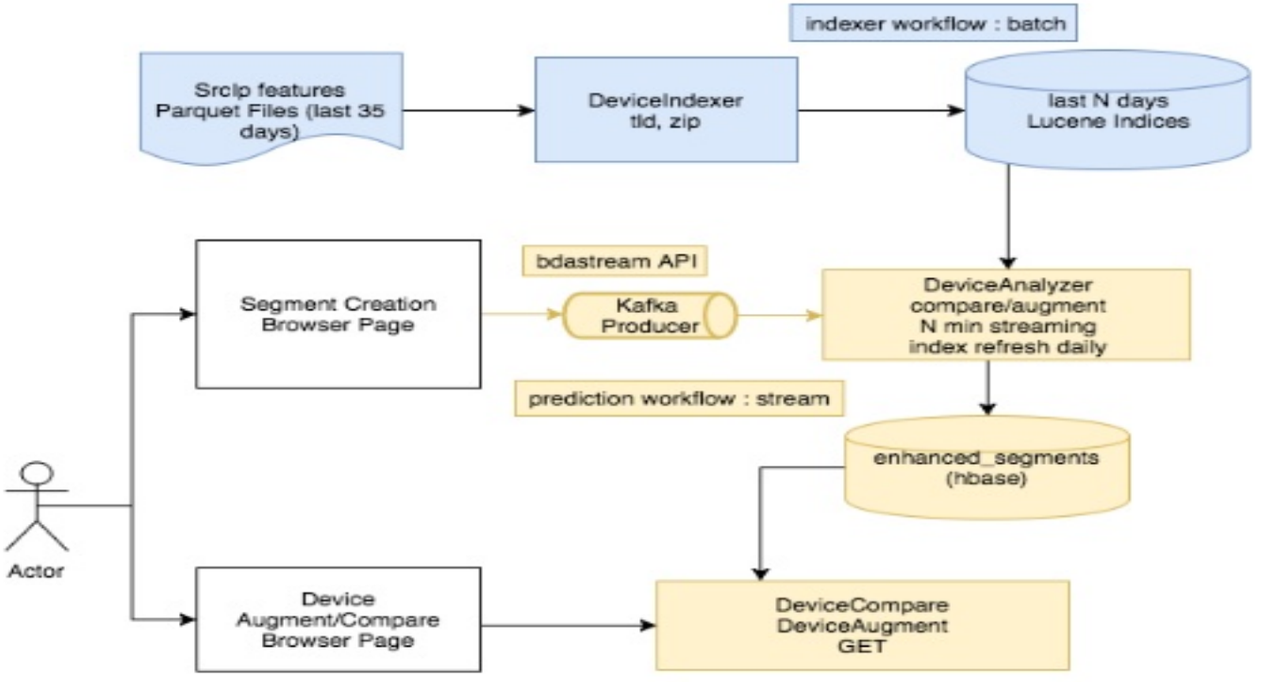
Trapezium Stream

```
runMode = "STREAM"
dataSource = "KAFKA"
kafkaTopicInfo = {
  consumerGroup = "KafkaStreamGroup"
  maxRatePerPartition = 970
  batchTime = "5"
  streamsInfo = [{
    name = "queries"
    topicName = "deviceanalyzer"
}]
}
transactions = [{
  transactionName = DeviceAnalyzer"
  inputStreams = [{name: "queries"}]
  persistStreamName = "deviceanalyzer"
  isPersist = "true"
}]
```



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DeviceAnalyzer: Compare

- Given two queries

select * from Devices where tld='macys.com' OR 'nordstorm.com' AND (city='SanFrancisco') AND (device='Android')

select * from Devices where tld='macys.com' OR 'nordstorm.com' AND (city='Brussels') AND (device='Android')

- Find the dimensions that discriminate the devices associated with two groups

```
def processStream(streams: Map[String, DStream[Row]], workflowTime: Time): {
 streams("queries").collect().map{ requests =>
   group1 = dao.search(requests(0))
   group2 = dao.search(requests(1))
   response = runLDA(aud1, aud2, dict)
}
```

- Sparse weighted least squares using Breeze QuadraticMinimizer
- L1 Regularized logistic regression

```
def persistStream(responses: RDD[Row], batchTime: Time) {
    HBaseDAO.write(responses)
}
```


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DeviceAnalyzer: Augment

- Given a query

select * from Devices where
tId='macys.com' OR 'nordstorm.com'
AND (city='SanFrancisco' OR 'Brussels')
AND (device='Android')...

  - Find devices similar to seed as lookalikes
  - Find dimensions that represent lookalikes

```
object DeviceAnalyzer extends StreamingTransaction {
converter = SparkLuceneConverter(dm.size)
batchTime = Trapezium.getSyncTime("indexer")
dao = LuceneDAO(batchTime...)
    .setConverter(converter).load(sc, indexPath)
dict = loadDictionary(sc, indexPath, batchTime)
all = dao.search("*:*")
def processStream(streams: Map[String, DStream[Row]]) :
{
  streams("queries").collect().map{ request =>
    audience = dao.search(request)
    response = getLookalikeDimensions(all, audience, dict)
}
```

- Sparse weighted least squares using Breeze QuadraticMinimizer
- L2 regularized linear regression

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FastSummarizer

- Statistical and predictive operators
  - sum: sum over numeric measures
  - support: sum over distinct docID
  - sumSquared: L2 norm
  - gram: Uses BLAS sspr
  - solve: Uses BreezeQuadraticMinimizer to support L1
- Implemented using Array[Float] for shuffle opt
- Scala/Java for Level1 operations
- OpenBLAS for Level3 operations

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Sync API Benchmark

73M rows 1M+ search terms

1 measure on 250K sparse dimensions

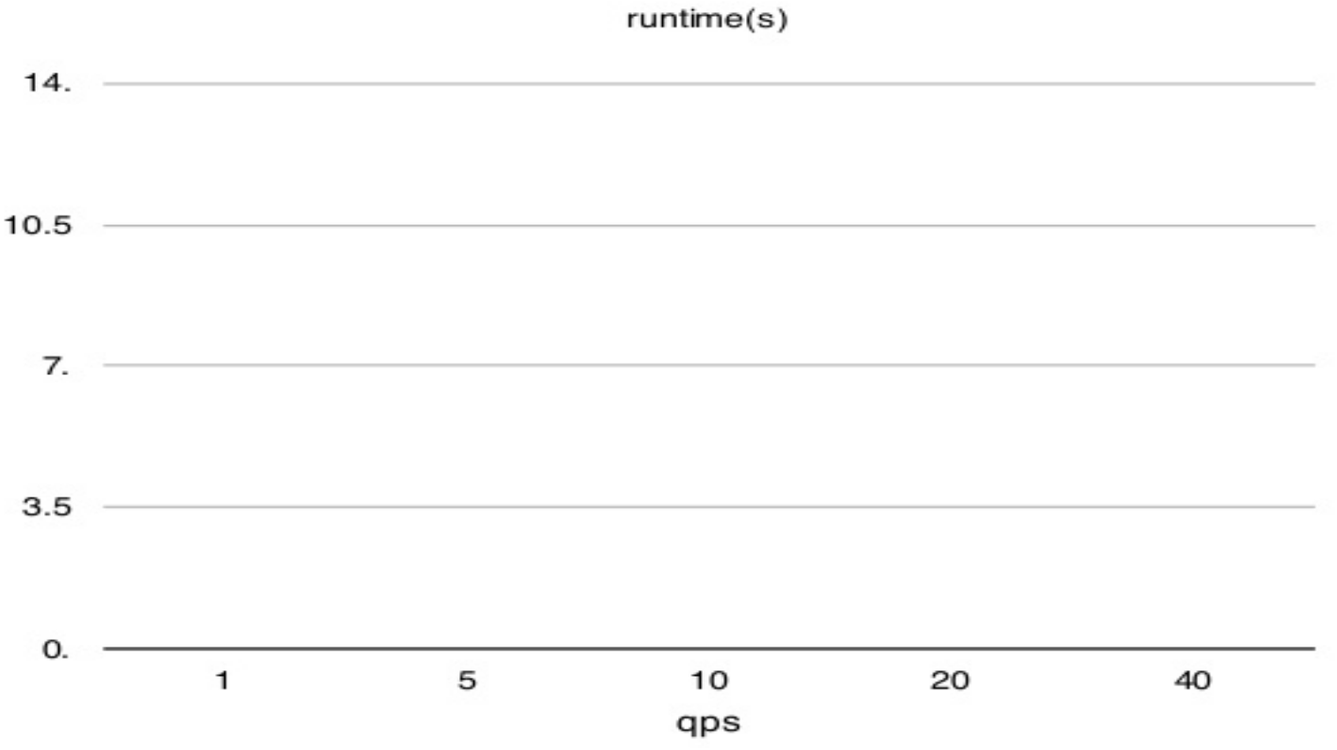
20 executors 8 cores

32 GB driver RAM 16 GB executor RAM

akka-http cores: 24 default

topk

qps	runtime(s)
1	1.389
5	1.663
10	3.214
20	5.992
40	12.174



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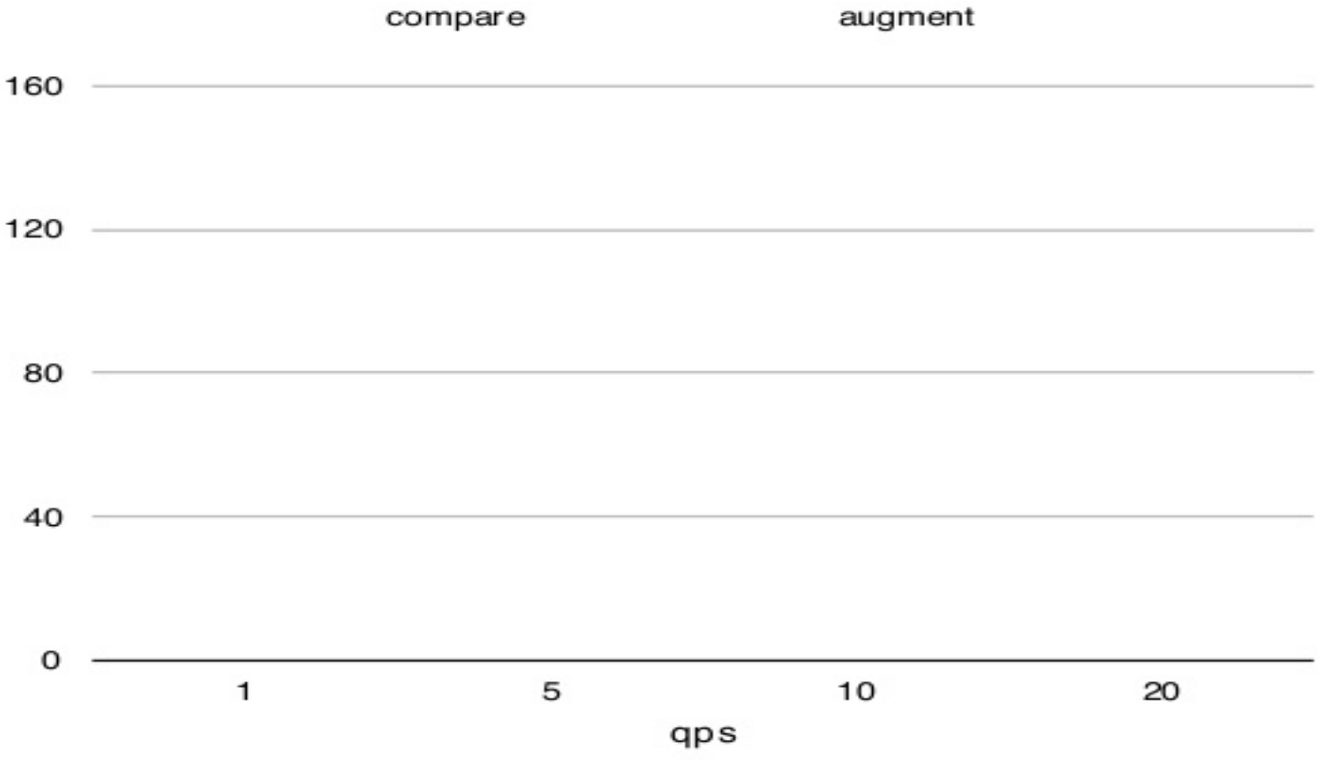
Async API Benchmark

73M rows, 1M+ search terms
1 measure on 250K sparse dimensions
20 executors 8 cores
32 GB driver RAM 16 GB executor RAM
forkjoinpool = 40
Kafka Fetch + compare/augment + HBase Persist

predictions		
qps	**compare(s)**	**augment(s)**
1	9	16
5	13	36
10	23	70
20	42	142



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topk tld + apps

Behaviors | Search for Sites, Apps and Keywords | Audience Definition | Filter Audience

Behaviors
Demographics
Devices
Locations

foxnews.com | npr.org

Behaviors

Rank	Site	Audience %
31	realtor.com	5.58%
32	usatoday.com	5.33%
33	stumbleupon.com	5.16%
34	disney.com	5.11%
35	zillow.com	5.04%
36	dailymail.co.uk	5.01%
37	starwars.com	4.85%
38	latimes.com	4.83%
39	truste.com	4.75%
40	nypost.com	4.7%

Previous 1 2 3 4 5 Next

Rank	App	Audience %
11	com.apple.mobilesafari	2.58%
12	Find My Friends	1.39%
13	Instagram	1.28%
14	com.zillow.zillowmap	1.21%
15	Starbucks	0.84%
16	Find My iPhone	0.72%
17	com.acmeaom.android.radarym	0.72%
18	com.facebook.katana	0.7%
19	com.apple.geod	0.66%
20	com.hearst.kcraiphone	0.63%

Previous 1 2 3 4 5 Next

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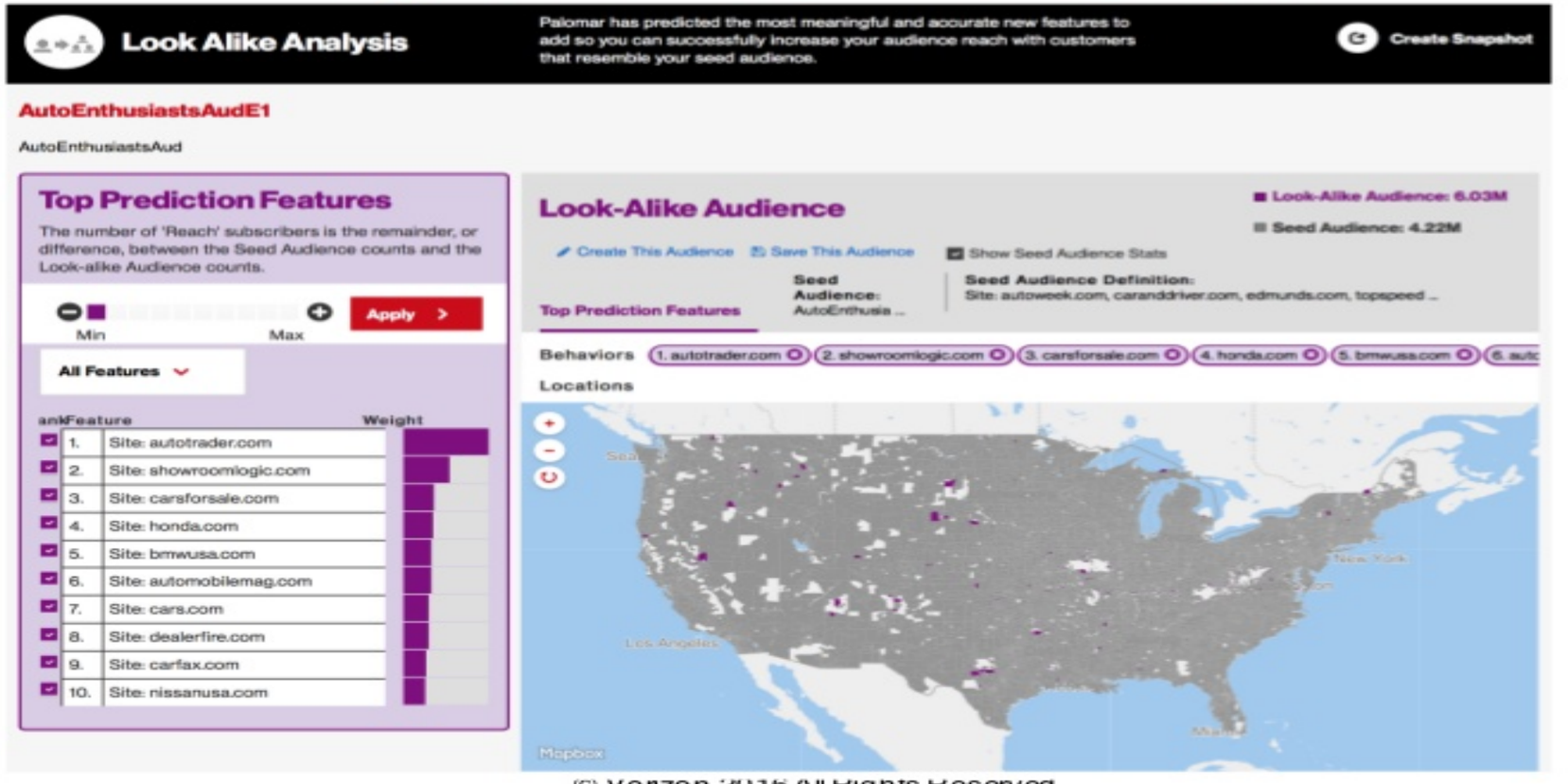
Augment: Auto Enthusiastic



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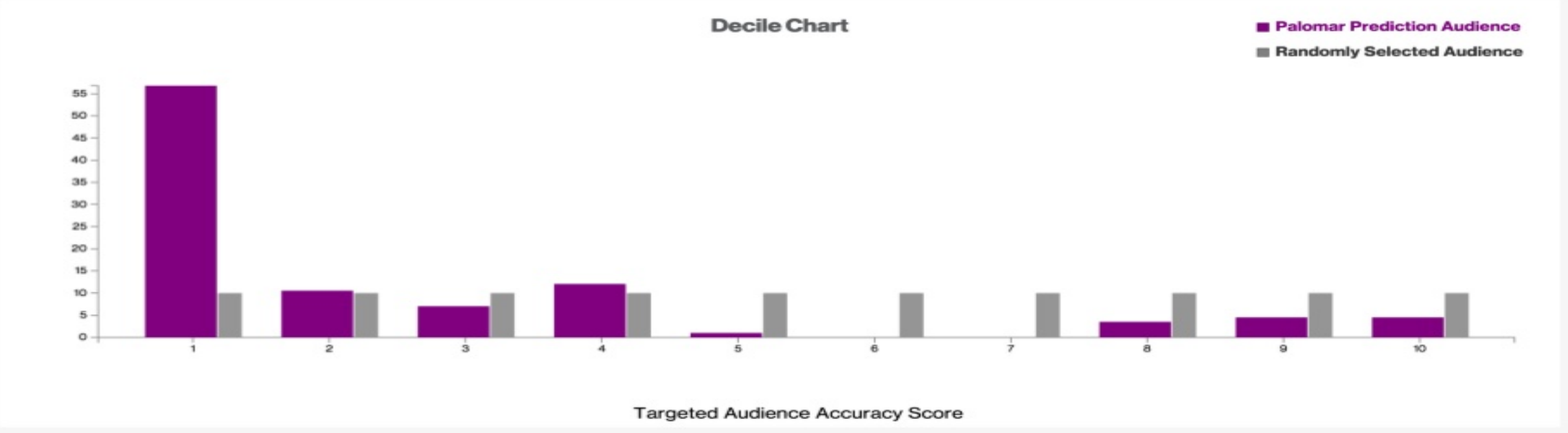
Augment Model Performance



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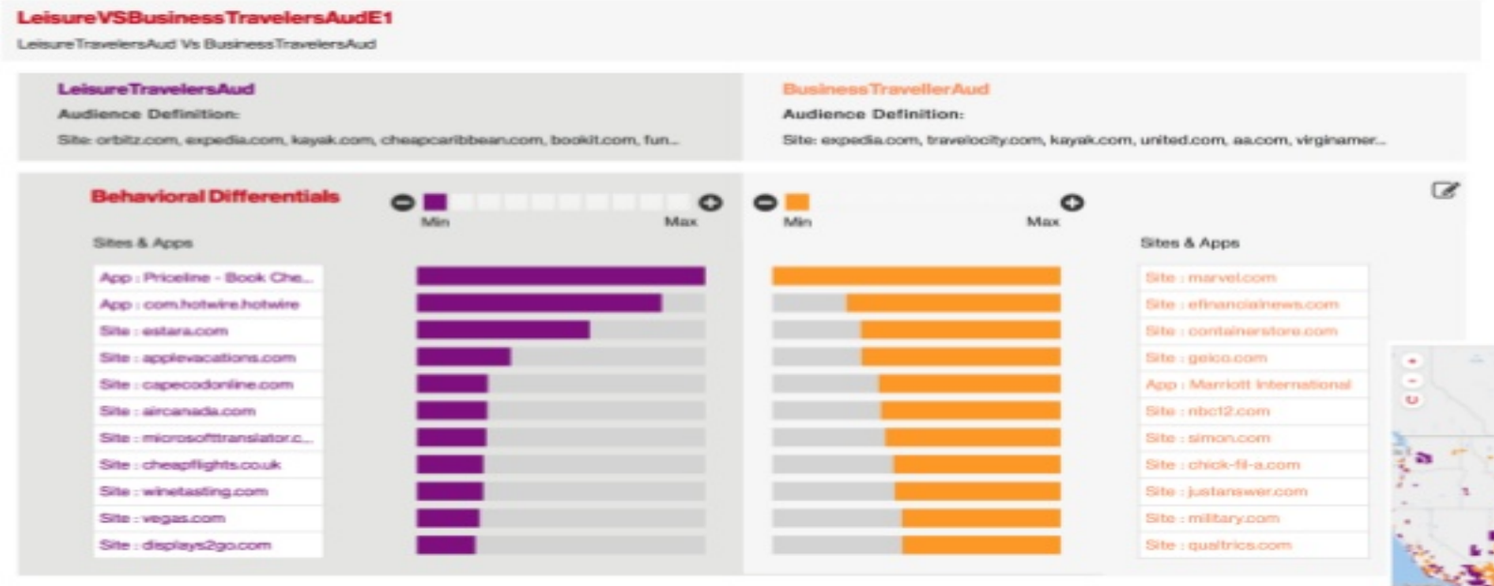
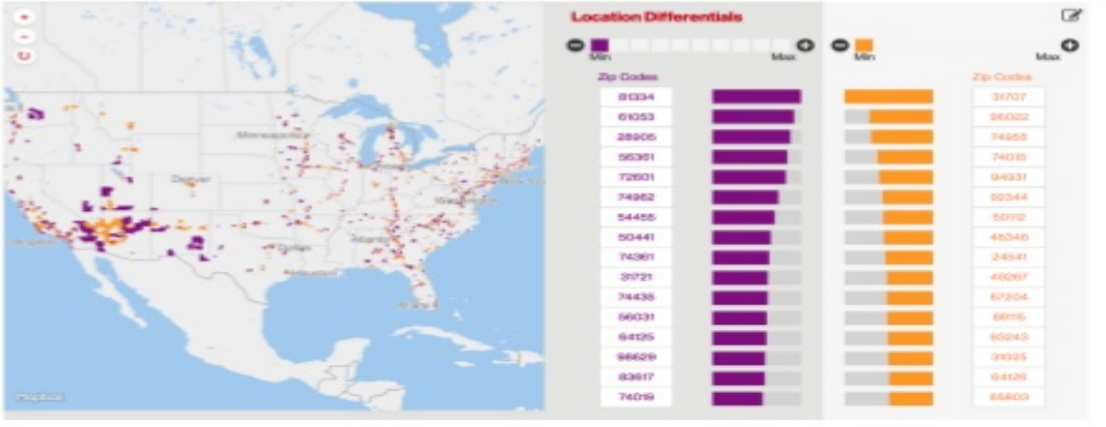
Compare: Leisure vs Business Travellers





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